

CASEY STATION, Antarctica

WMO: 896110

Lat: 66.282S

Lon: 110.523E

Elev: 42

StdP: 100.82

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	-29.0	-27.1	-33.2	0.2	-28.3	-31.4	0.2	-26.8	39.6	-8.7	37.1	-9.6	6.2	170	1.214

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.4	3.4	0.4	2.6	0.0	1.8	-0.5	0.9	2.7	0.3	2.0	-0.2	1.4	7.1	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
-0.7	3.6	0.6	-1.3	3.4	0.4	-1.9	3.2	0.2	11.3	2.9	10.4	2.2	9.4	1.5	3.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 33.5	29.4	25.0	-31.3	5.4	2.0	1.2	-32.7	6.2	-33.8	6.9	-34.9	7.6	-36.4	8.4		
(5)			-31.5	1.8	1.9	0.8	-32.8	2.4	-34.0	2.8	-35.1	3.2	-36.5	3.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-9.5	-0.5	-2.9	-7.3	-11.5	-14.6	-14.6	-14.8	-15.0	-13.8	-11.9	-5.6	-1.4	(6)
(7)		DBStd	7.09	1.70	2.41	3.69	5.07	6.08	5.91	6.17	6.00	5.37	4.13	3.15	1.94	(7)
(8)		HDD10.0	7122	325	362	536	646	762	737	768	774	713	678	467	353	(8)
(9)		HDD18.3	10164	583	595	794	896	1020	987	1027	1033	963	936	717	612	(9)
(10)		CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(10)
(11)		CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	7.1	5.4	6.0	6.8	7.5	7.2	7.8	8.6	8.2	8.4	7.0	7.0	5.9	(14)
(15)	Precipitation	PrecAvg	457	39	40	41	36	32	39	45	45	35	31	39	37	(15)
(16)		PrecMax	587	55	67	72	70	53	69	79	77	61	50	70	70	(16)
(17)		PrecMin	362	21	5	19	13	7	12	20	17	15	11	10	15	(17)
(18)		PrecStd	56	10	16	13	14	11	13	17	14	11	10	14	11	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.7	3.3	0.8	-0.5	-0.9	-2.3	-1.1	-1.0	-1.3	-2.0	1.9	4.5	(19)
(20)			MCWB	1.2	0.5	-1.3	-2.0	-2.1	-3.4	-2.2	-1.9	-2.4	-3.6	-0.9	0.9	(20)
(21)		2%	DB	3.7	2.2	-0.5	-2.0	-2.5	-3.9	-3.0	-3.2	-3.7	-3.4	0.3	3.1	(21)
(22)			MCWB	0.6	-0.2	-2.4	-3.4	-3.5	-4.9	-3.8	-4.1	-4.9	-4.8	-1.8	0.2	(22)
(23)		5%	DB	2.9	1.3	-1.4	-3.4	-4.3	-5.1	-4.8	-4.8	-5.1	-4.6	-0.6	2.2	(23)
(24)			MCWB	0.2	-0.8	-3.0	-4.7	-5.2	-6.1	-5.8	-5.7	-6.3	-5.8	-2.4	-0.3	(24)
(25)		10%	DB	2.2	0.4	-2.4	-4.7	-6.0	-6.4	-6.4	-6.6	-6.5	-5.7	-1.4	1.4	(25)
(26)			MCWB	-0.2	-1.4	-3.7	-5.8	-6.8	-7.3	-7.3	-7.5	-7.5	-6.8	-3.0	-0.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.7	1.1	-0.9	-1.7	-1.7	-3.0	-1.8	-1.7	-2.1	-3.2	-0.3	1.4	(27)
(28)			MCD B	3.9	2.7	0.2	-0.9	-0.9	-2.5	-1.5	-1.0	-1.6	-2.5	0.9	3.6	(28)
(29)		2%	WB	1.0	0.2	-2.0	-3.2	-3.5	-4.8	-3.7	-3.9	-4.7	-4.7	-1.4	0.6	(29)
(30)			MCD B	3.0	1.7	-0.8	-2.0	-2.7	-4.0	-3.1	-3.2	-3.9	-3.6	-0.1	2.4	(30)
(31)		5%	WB	0.6	-0.5	-2.9	-4.6	-5.2	-6.0	-5.8	-5.6	-6.2	-5.7	-2.2	0.0	(31)
(32)			MCD B	2.3	0.9	-1.7	-3.6	-4.3	-5.2	-4.9	-4.9	-5.3	-4.6	-0.8	1.7	(32)
(33)		10%	WB	0.1	-1.3	-3.7	-5.7	-6.8	-7.2	-7.3	-7.5	-7.4	-6.8	-2.9	-0.5	(33)
(34)			MCD B	1.8	0.2	-2.5	-4.8	-6.0	-6.5	-6.4	-6.7	-6.5	-5.8	-1.5	1.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	3.4	3.7	4.4	5.3	5.6	5.7	6.0	5.7	5.4	5.2	4.7	3.7	(35)
(36)			MCDBR	4.3	4.2	4.3	5.3	6.2	5.6	6.0	5.9	5.2	4.8	4.8	4.3	(36)
(37)			MCWBR	2.8	3.1	3.5	4.8	5.6	5.1	5.4	5.4	4.6	4.1	3.8	2.9	(37)
(38)		5% WB	MCDBR	3.7	3.8	4.2	5.2	6.4	5.7	5.9	6.0	5.2	4.7	4.5	3.8	(38)
(39)			MCWBR	2.5	2.8	3.5	5.0	6.1	5.4	5.5	5.7	4.8	4.1	3.7	2.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.267	0.256	0.234	0.194	0.129	0.129	0.134	0.191	0.221	0.230	0.251	0.269	(40)	
(41)		taud	2.344	2.384	2.365	2.224	1.925	1.925	1.898	2.167	2.272	2.287	2.298	2.290	(41)	
(42)		Ebn at Noon	995	953	878	705	364	66	310	687	874	976	1008	1006	(42)	
(43)		Edh at Noon	108	90	70	43	14	3	13	44	74	98	112	118	(43)	
(44)	All-Sky Solar Radiation	RadAvg	7.70	5.34	2.81	0.96	0.17	0.01	0.07	0.58	2.11	4.73	7.36	8.63	(44)	
(45)		RadStd	0.34	0.29	0.15	0.07	0.01	0.00	0.00	0.04	0.10	0.14	0.26	0.29	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page